

CASE STUDY: DRIVING 20% PRODUCTION EFFICIENCY

Transforming KPI Frameworks from Activity-Based to Outcome-Driven

Executive Summary

Successfully addressed systemic delivery delays by redesigning production performance metrics. By shifting the focus from "Daily Units Produced" to "Floors Completed and Shipped," the organization improved production efficiency by ~20%, significantly reduced backlogs, and aligned cross-functional teams with actual project delivery goals.

1. CONTEXT & CHALLENGE

The production facility was reporting high daily activity levels, yet project delivery timelines were consistently missed. The existing KPI system tracked "Units Produced," creating a "High Activity, Low Completion" environment.

- **The Problem:** Teams were optimizing for individual department targets rather than the finished product.
- **The Impact:** Growing backlogs, delayed site deliveries, and a disconnect between production, planning, and shipping departments.

2. STRATEGIC APPROACH

I treated this as a **Performance Management System failure** rather than a technical production issue.

- **Value Stream Mapping (VSM):** Analyzed the end-to-end flow to identify where "completed units" were stalling in the system.
- **Lean Principles:** Identified non-value-added activities that contributed to high activity counts but zero progress toward project completion.
- **Stakeholder Alignment:** Facilitated workshops between Production, Planning, and Logistics to define what a "Success Metric" actually looks like for the business.

3. THE SOLUTION: KPI REDESIGN

I overhauled the performance structure to prioritize **Flow and Completion**:

- **New Metric:** Transitioned from *Units Produced* to *Floors Completed and Shipped*.
- **Outcome-Based Execution:** Linked production rewards and schedules to the delivery of installable sets rather than individual parts.
- **Cross-Functional Sync:** Implemented a shared dashboard for all departments to track progress toward the same "Completion" goal.

4. QUANTIFIED RESULTS (ROI)

Key Metric	Baseline	Post-Intervention	Improvement
Production Efficiency	Fragmented	Synchronized	~20% Increase
Backlog Volume	High/Increasing	Significantly Reduced	Major Reduction
Delivery Performance	Unpredictable	On-Time / Project Aligned	Tier-1 Level

5. LEADERSHIP TAKEAWAYS

- **KPIs Drive Behavior:** If you measure activity, you get busyness. If you measure outcomes, you get results.
- **Systems Over Silos:** Efficiency is found in the *transition* between departments, not just within them.
- **Lean Governance:** Reducing waste is as much about shifting mindset as it is about shifting processes.

Prepared by: Hussain Syed | P.Eng, PMP

Expertise: Lean Manufacturing & Operational Excellence